

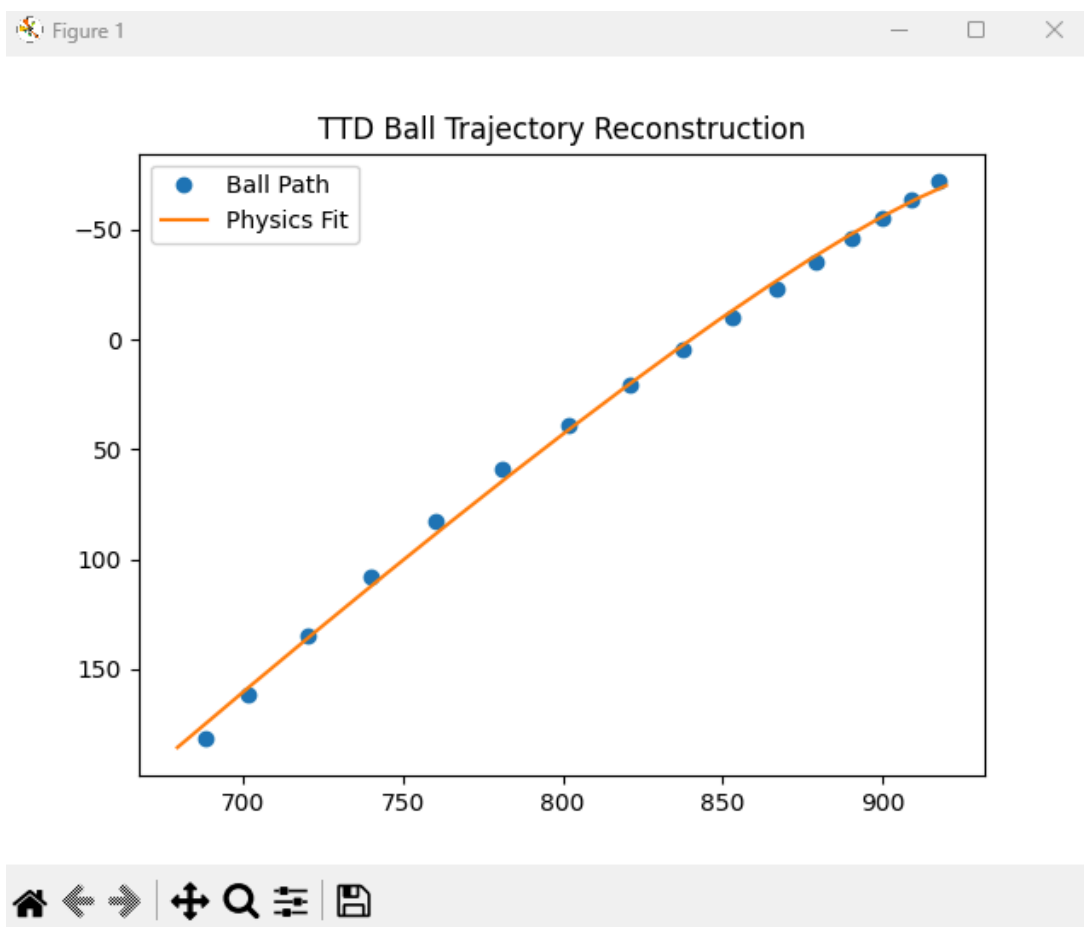
Basketball Trajectory Reconstruction ML Pipeline

Computer vision system for trajectory extraction and physics-based motion reconstruction

Demo: <https://youtube.com/watch?v=luLLZKf26uw>

GitHub: <https://github.com/msuero-alt/basketball-shot-trajectory>

An end-to-end machine learning pipeline for vision-based object detection and physics-informed trajectory reconstruction from video.



- Input video is processed frame-by-frame using a YOLOv8 object detection model
- Ball coordinates are extracted from detections and structured into a time-series sequence

Technologies: Python, OpenCV, YOLOv8, NumPy, Matplotlib

Detection & Tracking Pipeline

```
=== TTD DEMO START ===  
RUN BALL STARTED  
LOCKED ON BALL  
Frame 0: BALL (688.2420043945312, 181.22665405273438)  
Frame 1: BALL (701.527099609375, 161.4970703125)  
Frame 2: BALL (720.1094970703125, 134.42654418945312)  
Frame 3: BALL (739.866455078125, 107.59489440917969)  
Frame 4: BALL (760.5123291015625, 82.55982971191406)  
Frame 5: BALL (781.081787109375, 58.94763946533203)  
Frame 6: BALL (801.9982299804688, 38.86080551147461)  
Frame 7: PREDICTED (820.822998046875, 20.782655715942383)  
Frame 8: PREDICTED (837.7653198242188, 4.512321472167969)  
Frame 9: PREDICTED (853.013427734375, -10.13097858428955)  
Frame 10: PREDICTED (866.7366943359375, -23.309947967529297)  
Frame 11: PREDICTED (879.087646484375, -35.1710205078125)  
Frame 12: PREDICTED (890.2034912109375, -45.845985412597656)  
Frame 13: PREDICTED (900.207763671875, -55.453453063964844)  
Frame 14: PREDICTED (909.2116088867188, -64.10017395019531)  
Frame 15: PREDICTED (917.3150634765625, -71.8822250366211)  
FINAL COORD COUNT: 16  
  
Total tracked points: 16  
  
=== SHOT SUMMARY ===  
Frames tracked: 16  
Estimated gravity (relative): 1.5698
```

The system logs frame-level detections and transitions from active tracking to predictive modeling once sufficient motion data is collected.

System Pipeline Overview

```
# 1. LOAD MODEL  
model = YOLO("yolov8n.pt")  
  
# 2. TRACK BALL  
coords = detect_ball(VIDEO_PATH, model)  
  
coords = np.array(coords)  
  
print(f"\nTotal tracked points: {len(coords)}")  
  
if len(coords) < 5:  
    print("Not enough data for shot reconstruction.")  
    exit()  
  
# 3. PHYSICS FIT  
t, x_curve, y_curve = fit_projectile(coords)  
  
g_est = estimate_gravity(y_curve)
```

End-to-end machine learning pipeline combining vision-based detection with physics-based trajectory reconstruction.